Modernizing an app-based Aphasia Rehabilitation Tool

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My Background:

- **Bachelor of Science, 2024**
  - **Major:** Computer Science
  - **Minor:** Psychology

- **Master of Science, 2026**
  - Computer Science, Cognitive Science
- **PhD, Computer Science, TBD**

Research Interests: Human-Computer Interaction (HCI), Brain-Computer Interfaces (BCI), HCI for disability

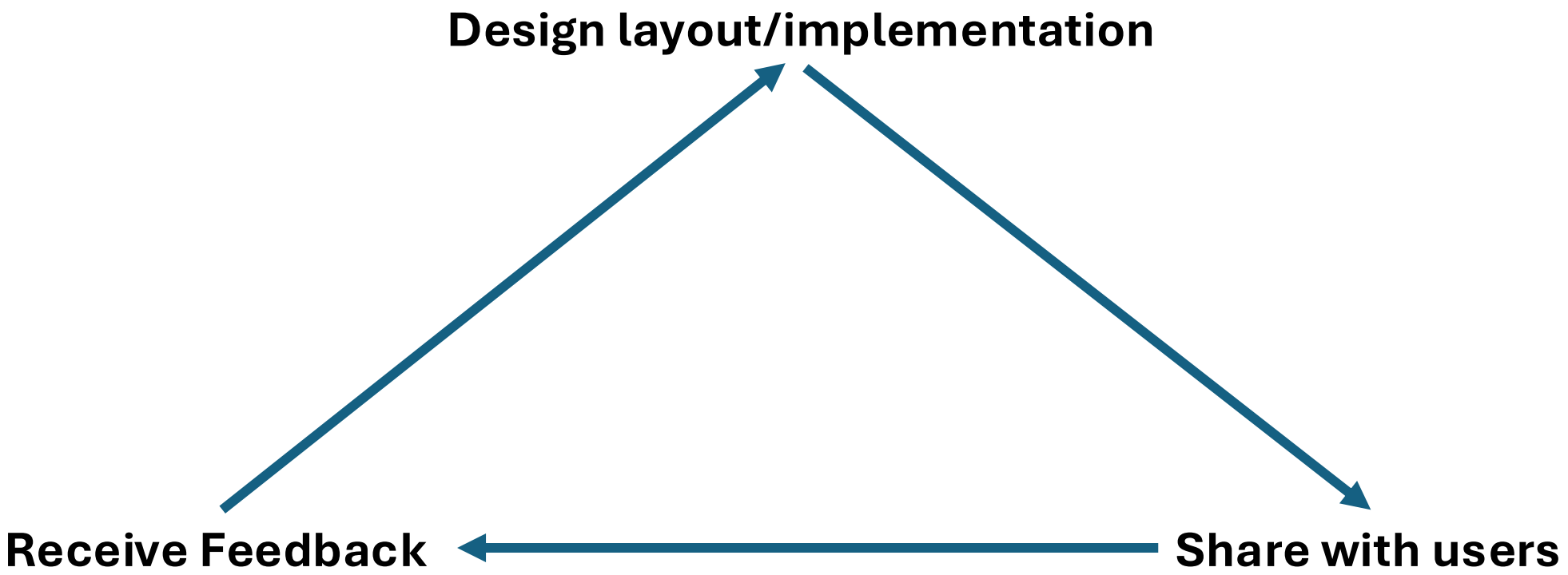
Human-Computer Interaction

- **Brain-Computer Interfaces:** using brain measurements to enhance user experience
  - fNIRS, EEG
- **Ability-based design:** An approach to digital accessibility that focuses on *adapting systems to users' abilities* rather than forcing users to adapt to technology (Wobbrock et al., 2011)
  - **Adaptivity** – tech adjusting to user
  - **Adaptability** – customizability to users

  Codesign – incorporating people with disabilities in the design process



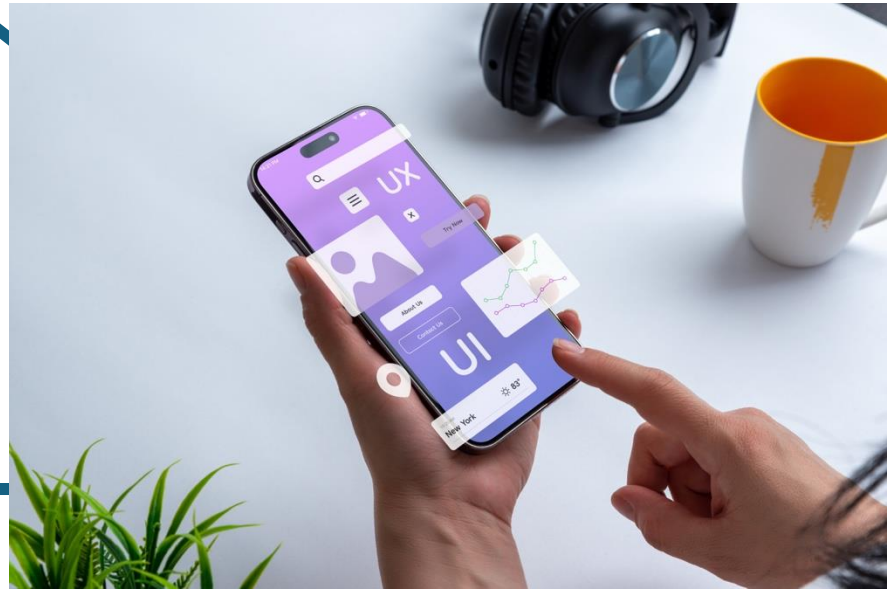
HCI and Codesign



HCI and Codesign

Design layout/implementation





HCI and Codesign



Receive Feedback ← **Share with users**

HCI and Codesign



Receive Feedback ← **Share with users**

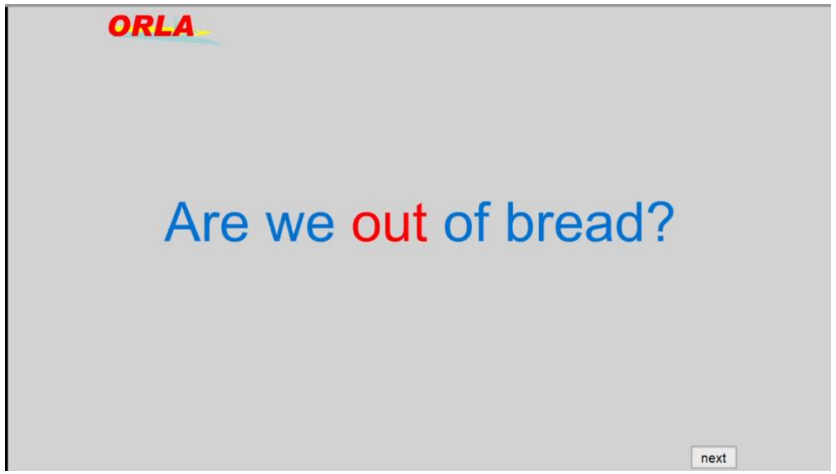
ORLA

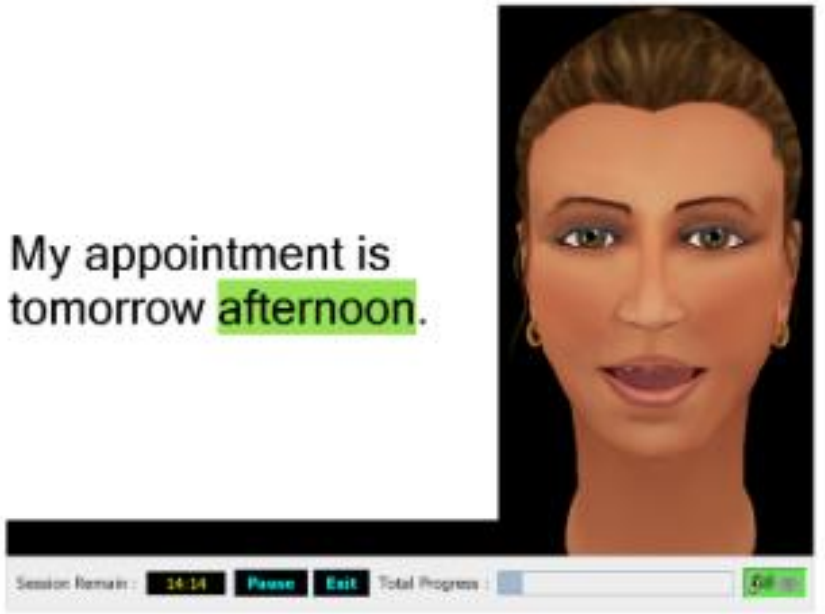
During a session, the person with aphasia repeatedly reads sentences or paragraphs aloud:

- **Look and Listen:** Clinician reads the text aloud while pointing to each word
- **Joint Pointing:** Clinician reads aloud again, patient points
- **Choral Reading:** Clinician and patient read and point together
- **Word Identification:** Clinician states a word in the sentence, patient points
- **Word Production:** Clinician points to a word, patient reads aloud independently.
- **Final Choral Reading:** Patient and clinician read the entire text aloud in unison.

The program is also graded into four levels of difficulty based on text length and reading level, allowing the clinician to adjust for a range of aphasia severities. (Cherney, 2010)

ORLA


ORLA
Are we out of bread?
next


My appointment is tomorrow afternoon.
Session Remain :
14:14
Pause
Exit
Total Progress :

Initial design

Framework: PyQt
Python framework for GUI creation

Audio: edge-tts, computationally generated

Visual aid: *Unsplash*, stock photos related to sentence



ORLA 2.0: Modernizing an App-Based Aphasia Rehabilitation Tool

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Figure 1: Comparison of the ORLA program currently in distribution (left) to the new ORLA beta (right).

ORLA (Oral Reading for Language in Aphasia):

systematic rehabilitation program that targets reading comprehension and other language modalities. (Cherney et al., 1986)

During an ORLA session, the person with aphasia repeatedly reads sentences or paragraphs aloud following these steps:

- **Look and Listen:** Clinician reads the text aloud while pointing to each word.
- **Joint Pointing:** Clinician reads aloud again, patient points
- **Choral Reading:** Clinician and patient read and point together
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References

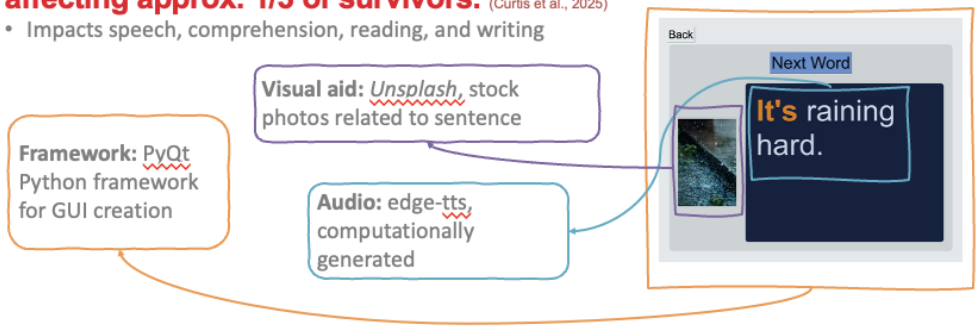
1. Cherney LR, Merbitz CT, Grip JC. Efficacy of oral reading in aphasia treatment outcome. Rehabil Lit. 1986 May-Jun;47(5-6):112-8. PMID: 3749589.
2. Cherney LR. Oral reading for language in aphasia: impact of aphasia severity on cross-modal outcomes in chronic nonfluent aphasia. Semin Speech Lang. 2010 Feb;31(1):42-51. doi: 10.1055/s-0029-1244952. Epub 2010 Apr 8. PMID: 20379961.
3. Curtis, H., Bircanin, F., & Neate, T. (2025). "An old bastard in bright orange satin!": Zuzenna's aphasia diary: And lessons learned from DIY augmentative and alternative communication. *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '25) (Article 48, pp. 1–20). Association for Computing Machinery. https://doi.org/10.1145/3663547.3746369.

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I would like to acknowledge my PhD advisor Robert Jacob, for his insight related to the technical and Human-Computer Interaction components of this project.

Aphasia: acquired language impairment, most caused by stroke, affecting approx. 1/3 of survivors. (Curtis et al., 2025)

- Impacts speech, comprehension, reading, and writing



Future Steps

Development is currently in beta. There are multiple future directions for this project:

- **Update text-to-speech:** edge-tts is robotic, update AI model for natural speech
- **Allow personalization:** Give the user the ability to choose the voices that read sentences, change theme, etc.
- **Gamification:** Increase motivation by adding features that ”gamify” the experience using a point system, intermittent reward, etc.

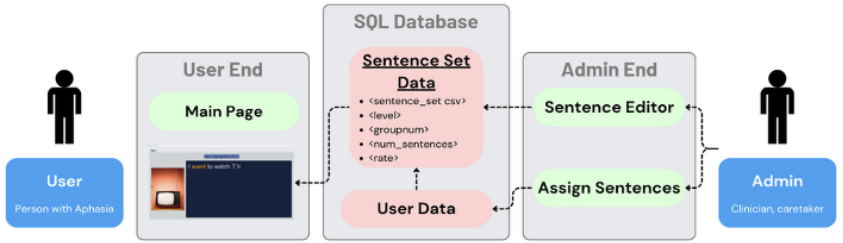


Figure 2. Data flow diagram showing the interface between admin and user. The admin creates sentence sets and assigns them to users, this data is stored in the backend database, and the user only sees the sentences without access to any of the data.

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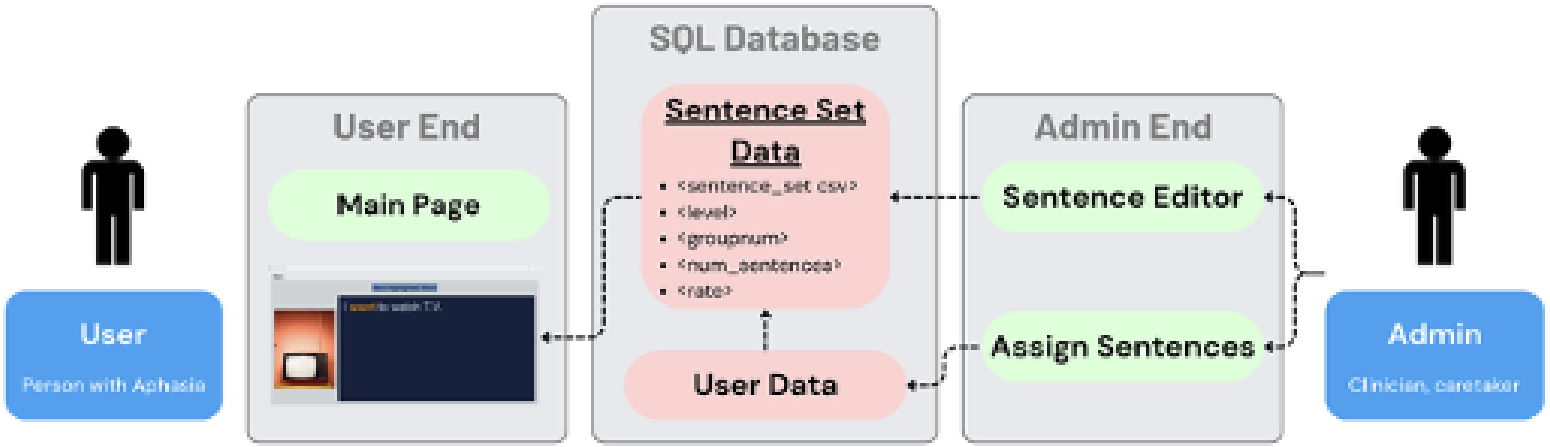
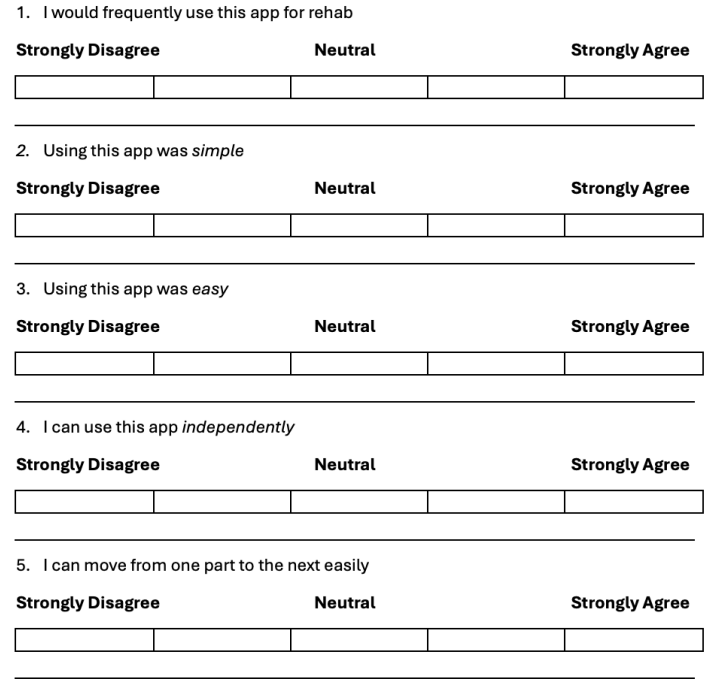


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Aphasia Design Interviews

- Started with simply having the participant use the app

- Adapting aphasia-friendly design interview questions:
  - Based on the **Systems Usability Scale (SUS)**
  - Printed out sheets
    - Participants can answer by **pointing**
    - Open-ended questions with **answer options**



1. I would frequently use this app for rehab

Strongly Disagree		Neutral		Strongly Agree

2. Using this app was *simple*

Strongly Disagree		Neutral		Strongly Agree

3. Using this app was *easy*

Strongly Disagree		Neutral		Strongly Agree

4. I can use this app *independently*

Strongly Disagree		Neutral		Strongly Agree

5. I can move from one part to the next easily

Strongly Disagree		Neutral		Strongly Agree

Open-ended questions:

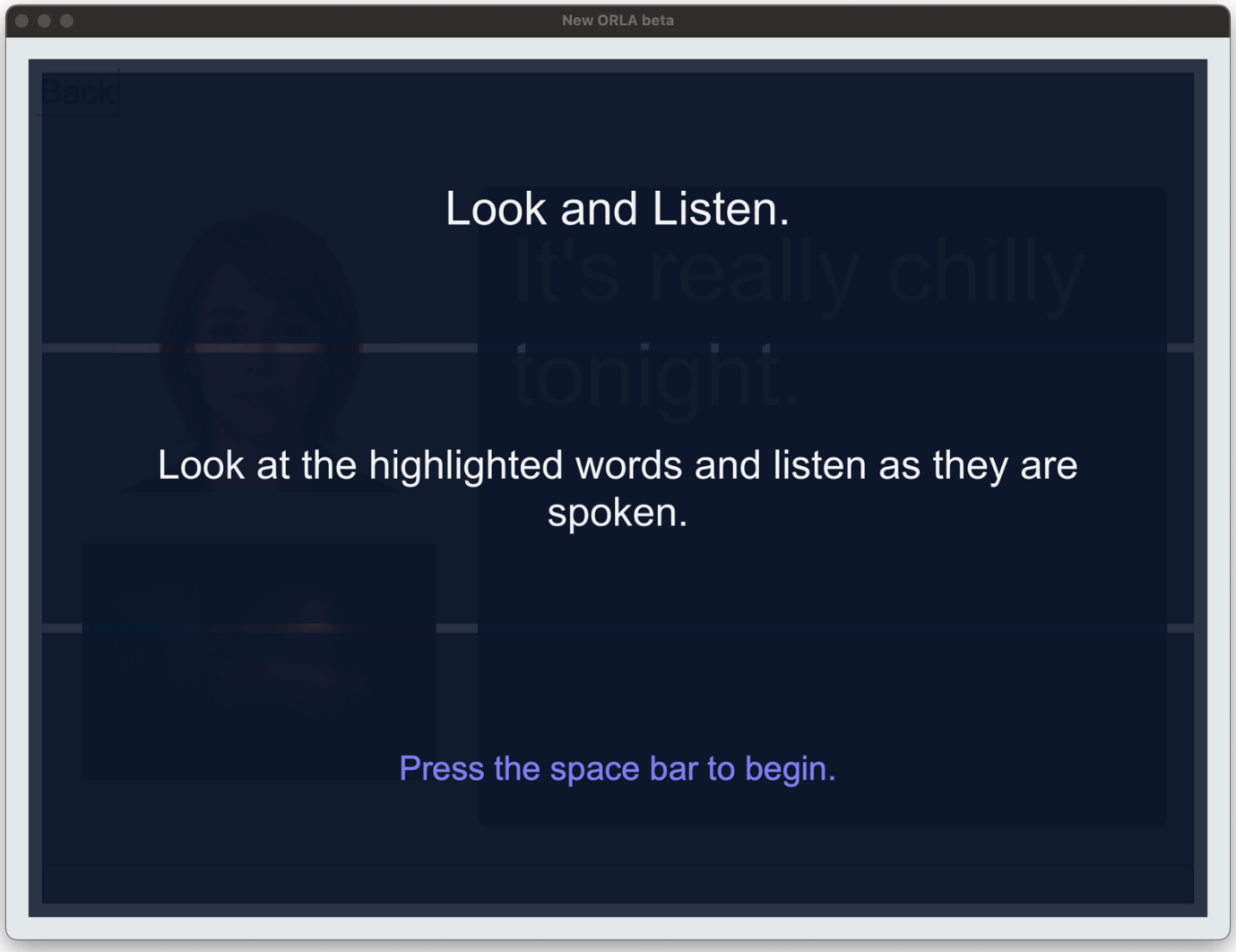
1. *Was there anything you thought was **missing** from this app?*
   a. *Nothing was missing*
   b. *Colors/images*
   c. *Voices*
   d. *Interaction (i.e. pressing buttons)*
   e. *Something else?*
2. *How did you feel about the **images** provided along with the sentences?*
   a. *Neutral*
   b. *Enjoyed them*
   c. *Didn't match the prompt well enough*
   d. *Distracting*
   e. *Something else?*

Aphasia Design Interviews: Debrief

- **2 community members interviewed**
  - Higher fluency level
- **The app was much buggier than expected**
  - Mostly observation, less questions
- **Not an evaluation of *them* but of the *app***

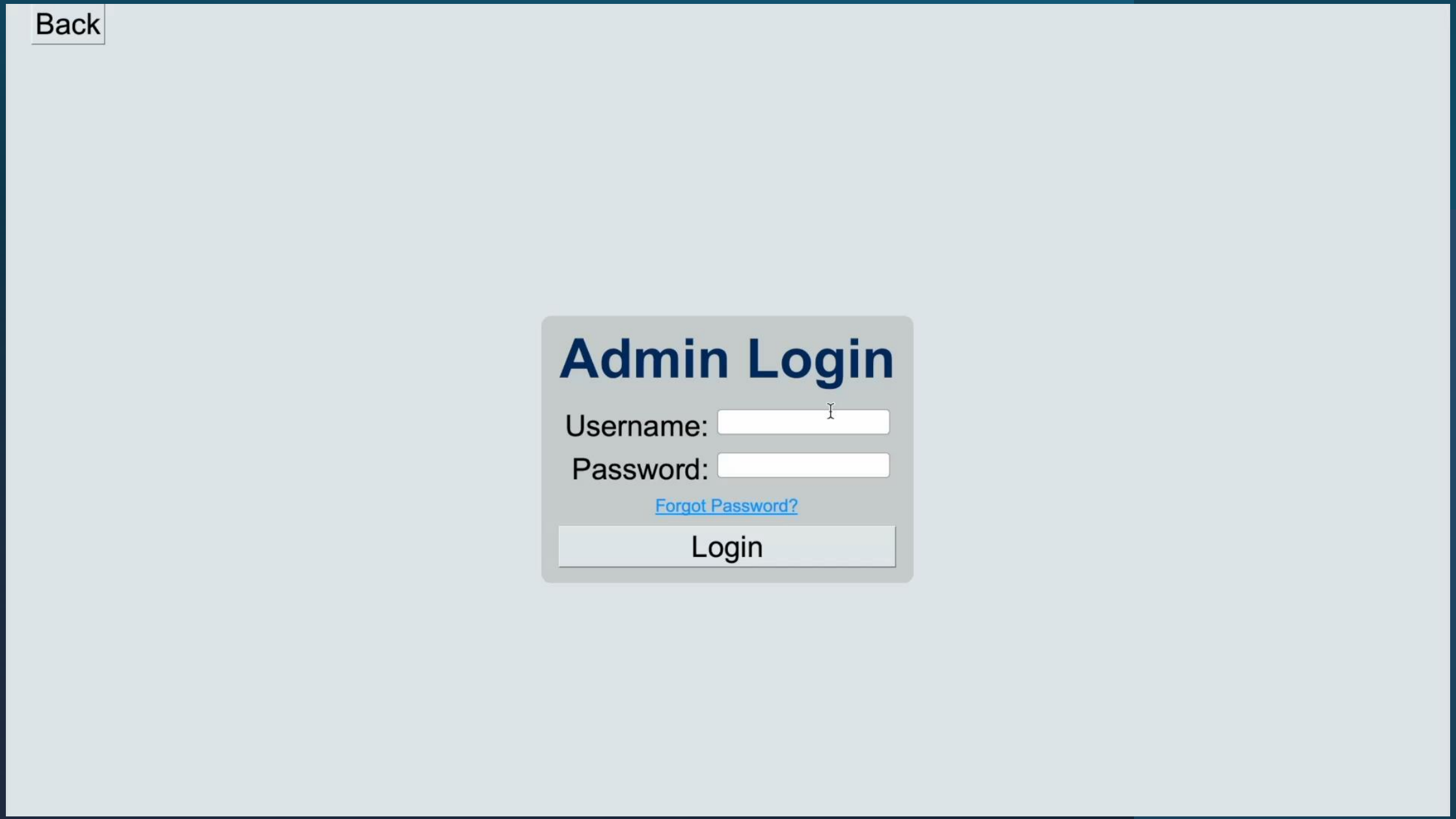
Aphasia Design Interviews: Changes Made

- **No *required* interactions**
  - Besides to advance section
  - Some *optional* interactions
- **Switched from edge-tts to Azure**
  - More speech customization



Additional Feature: Ms. PAT




Back
Admin Login
Username:
Password:
Forgot Password?
Login

Next Steps

- **General Debugging**
  - Password change
- **Admin features**
  - Write original sentences, upload sentence CSV’s
- **Gamification**
  - Minigames, unlocking avatars

